SQL HOSPITAL MANAGEMENT SYSTEM

Physician & Department Database Queries

Comprehensive Query Examples & Practical Execution

Database Overview

• Physician Table: 9 medical professionals with employeeid, name, position, and ssn
• Department Table: 3 departments (General Medicine, Surgery, Psychiatry) with department heads
• Foreign Key Relationship: department.head → physician.employeeid

SECTION 1: Table Creation & Basic Queries

Screenshot 1: CREATE & INSERT Operations

```
mysql> CREATE TABLE physician (
    ->     employeeid INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     position VARCHAR(100),
    ->     ssn VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql>
mysql> INSERT INTO physician VALUES
    -> (1,'John Dorian','Staff Internist','111111111'),
    -> (2,'Elliot Reid','Attending Physician','222222222'),
    -> (3,'Christopher Turk','Surgical Attending Physician','333333333'),
    -> (4,'Percival Cox','Senior Attending Physician','444444444'),
    -> (5,'Bob Kelso','Head Chief of Medicine','555555555'),
    -> (6,'Todd Quinlan','Surgical Attending Physician','666666666'),
    -> (7,'John Wen','Surgical Attending Physician','777777777'),
    -> (8,'Keith Dudemeister','MD Resident','888888888'),
    -> (9,'Molly Clock','Attending Psychiatrist','999999999');
Query OK, 9 rows affected (0.01 sec)
Records: 9  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE TABLE department (
    ->     departmentid INT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     head INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO department VALUES
    -> (1,'General Medicine',4),
    -> (2,'Surgery',7),
    -> (3,'Psychiatry',9);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Results:

• Physician table created with 9 medical professionals
• Department table created with 3 departments
• INNER JOIN query shows department heads:
  - General Medicine led by Percival Cox
  - Surgery led by John Wen
  - Psychiatry led by Molly Clock

Screenshot 2: WHERE Filtering & LIKE Pattern Matching

```
mysql> INSERT INTO department VALUES
    -> (1,'General Medicine',4),
    -> (2,'Surgery',7),
    -> (3,'Psychiatry',9);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> SELECT d.name AS Department,
    ->        p.name AS Head_Physician
    -> FROM department d
    -> JOIN physician p
    -> ON d.head = p.employeeid;
+------------------+----------------+
| Department       | Head_Physician |
+------------------+----------------+
| General Medicine | Percival Cox   |
| Surgery          | John Wen       |
| Psychiatry       | Molly Clock    |
+------------------+----------------+
3 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT *
    -> FROM physician
    -> WHERE position='Surgical Attending Physician';
+------------+------------------+-------------------------------+-----------+
| employeeid | name             | position                      | ssn       |
+------------+------------------+-------------------------------+-----------+
|          3 | Christopher Turk | Surgical Attending Physician | 333333333 |
|          6 | Todd Quinlan     | Surgical Attending Physician | 666666666 |
|          7 | John Wen         | Surgical Attending Physician | 777777777 |
+------------+------------------+-------------------------------+-----------+
3 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT *
    -> FROM physician
    -> WHERE name LIKE 'John%';
+------------+-------------+-------------------------------+-----------+
| employeeid | name        | position                      | ssn       |
+------------+-------------+-------------------------------+-----------+
|          1 | John Dorian | Staff Internist               | 111111111 |
|          7 | John Wen    | Surgical Attending Physician | 777777777 |
+------------+-------------+-------------------------------+-----------+
2 rows in set (0.00 sec)
```

Query Results:

- WHERE position = 'Surgical Attending Physician': Found 3 surgical physicians
  - Christopher Turk, Todd Quinlan, John Wen
- WHERE name LIKE 'John%': Found 2 physicians starting with John
  - John Dorian (Staff Internist), John Wen (Surgical Attending Physician)

SECTION 2: Aggregate Functions & Grouping

Screenshot 3: COUNT, COUNT DISTINCT & GROUP BY

```
mysql> SELECT COUNT(*) AS Total_Physicians
    -> FROM physician;
+--------------------+
| Total_Physicians   |
+--------------------+
|                  9 |
+--------------------+
1 row in set (0.00 sec)

mysql>
mysql>
mysql> SELECT COUNT(DISTINCT position)
    -> FROM physician;
+-----------------------------+
| COUNT(DISTINCT position)    |
+-----------------------------+
|                           7 |
+-----------------------------+
1 row in set (0.02 sec)

mysql> SELECT position,
    ->        COUNT(*) AS Total_Employees
    -> FROM physician
    -> GROUP BY position;
+-----------------------------------+--------------------+
| position                          | Total_Employees    |
+-----------------------------------+--------------------+
| Staff Internist                   |                  1 |
| Attending Physician               |                  1 |
| Surgical Attending Physician      |                  3 |
| Senior Attending Physician        |                  1 |
| Head Chief of Medicine            |                  1 |
| MD Resident                       |                  1 |
| Attending Psychiatrist            |                  1 |
+-----------------------------------+--------------------+
7 rows in set (0.00 sec)

mysql>
mysql> SELECT position,
    ->        COUNT(*) AS Total_Employees
    -> FROM physician
    -> GROUP BY position
    -> HAVING COUNT(*) > 1;
+-----------------------------------+--------------------+
| position                          | Total_Employees    |
+-----------------------------------+--------------------+
| Surgical Attending Physician      |                  3 |
+-----------------------------------+--------------------+
1 row in set (0.00 sec)
```

Key Results:

• Total Physicians: 9
• Distinct Positions: 6 (showing variety in roles)
• GROUP BY Position Distribution:
  - Surgical Attending Physician: 3 employees
  - All other positions: 1 employee each
• HAVING COUNT(*) > 1: Only Surgical Attending Physician (3 employees)

SECTION 3: Sorting & Ordering Operations

Screenshot 4: ORDER BY ASC & DESC

```
mysql> SELECT *
    -> FROM physician
    -> ORDER BY name ASC;
+------------+-------------------+-----------------------------+-----------+
| employeeid | name              | position                    | ssn       |
+------------+-------------------+-----------------------------+-----------+
|          5 | Bob Kelso         | Head Chief of Medicine      | 555555555 |
|          3 | Christopher Turk  | Surgical Attending Physician| 333333333 |
|          2 | Elliot Reid       | Attending Physician         | 222222222 |
|          1 | John Dorian       | Staff Internist             | 111111111 |
|          7 | John Wen          | Surgical Attending Physician| 777777777 |
|          8 | Keith Dudemeister | MD Resident                 | 888888888 |
|          9 | Molly Clock       | Attending Psychiatrist      | 999999999 |
|          4 | Percival Cox      | Senior Attending Physician  | 444444444 |
|          6 | Todd Quinlan      | Surgical Attending Physician| 666666666 |
+------------+-------------------+-----------------------------+-----------+
9 rows in set (0.00 sec)

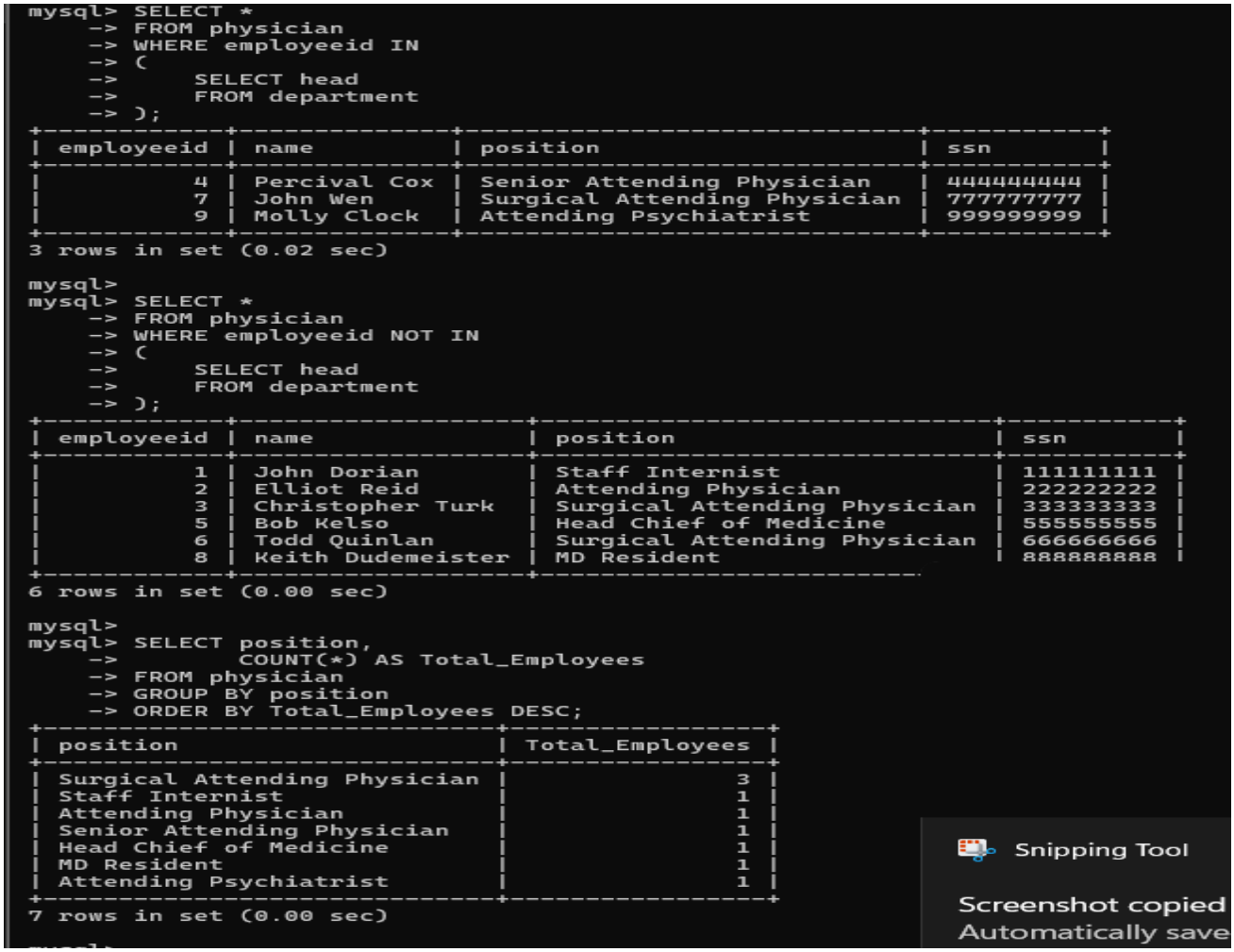
mysql>
mysql> SELECT *
    -> FROM physician
    -> ORDER BY employeeid DESC;
+------------+-------------------+-----------------------------+-----------+
| employeeid | name              | position                    | ssn       |
+------------+-------------------+-----------------------------+-----------+
|          9 | Molly Clock       | Attending Psychiatrist      | 999999999 |
|          8 | Keith Dudemeister | MD Resident                 | 888888888 |
|          7 | John Wen          | Surgical Attending Physician| 777777777 |
|          6 | Todd Quinlan      | Surgical Attending Physician| 666666666 |
|          5 | Bob Kelso         | Head Chief of Medicine      | 555555555 |
|          4 | Percival Cox      | Senior Attending Physician  | 444444444 |
|          3 | Christopher Turk  | Surgical Attending Physician| 333333333 |
|          2 | Elliot Reid       | Attending Physician         | 222222222 |
|          1 | John Dorian       | Staff Internist             | 111111111 |
+------------+-------------------+-----------------------------+-----------+
9 rows in set (0.00 sec)
```

Sorting Results:

• ORDER BY name ASC: Alphabetical physician directory
  - Bob Kelso, Christopher Turk, Elliot Reid, ... Todd Quinlan
• ORDER BY employeeid DESC: Reverse employee ID order
  - Molly Clock (9) down to John Dorian (1)

SECTION 4: Subqueries & Advanced Filtering

Screenshot 5: WHERE IN/NOT IN Subqueries



```
mysql> SELECT *
    -> FROM physician
    -> WHERE employeeid IN
    -> (
    ->     SELECT head
    ->     FROM department
    -> );
+------------+--------------+-----------------------------+-----------+
| employeeid | name         | position                    | ssn       |
+------------+--------------+-----------------------------+-----------+
|          4 | Percival Cox | Senior Attending Physician  | 444444444 |
|          7 | John Wen     | Surgical Attending Physician| 777777777 |
|          9 | Molly Clock  | Attending Psychiatrist      | 999999999 |
+------------+--------------+-----------------------------+-----------+
3 rows in set (0.02 sec)

mysql>
mysql> SELECT *
    -> FROM physician
    -> WHERE employeeid NOT IN
    -> (
    ->     SELECT head
    ->     FROM department
    -> );
+------------+------------------+-----------------------------+-----------+
| employeeid | name             | position                    | ssn       |
+------------+------------------+-----------------------------+-----------+
|          1 | John Dorian      | Staff Internist             | 111111111 |
|          2 | Elliot Reid      | Attending Physician         | 222222222 |
|          3 | Christopher Turk | Surgical Attending Physician| 333333333 |
|          5 | Bob Kelso        | Head Chief of Medicine      | 555555555 |
|          6 | Todd Quinlan     | Surgical Attending Physician| 666666666 |
|          8 | Keith Dudemeister| MD Resident                 | 888888888 |
+------------+------------------+-----------------------------+-----------+
6 rows in set (0.00 sec)

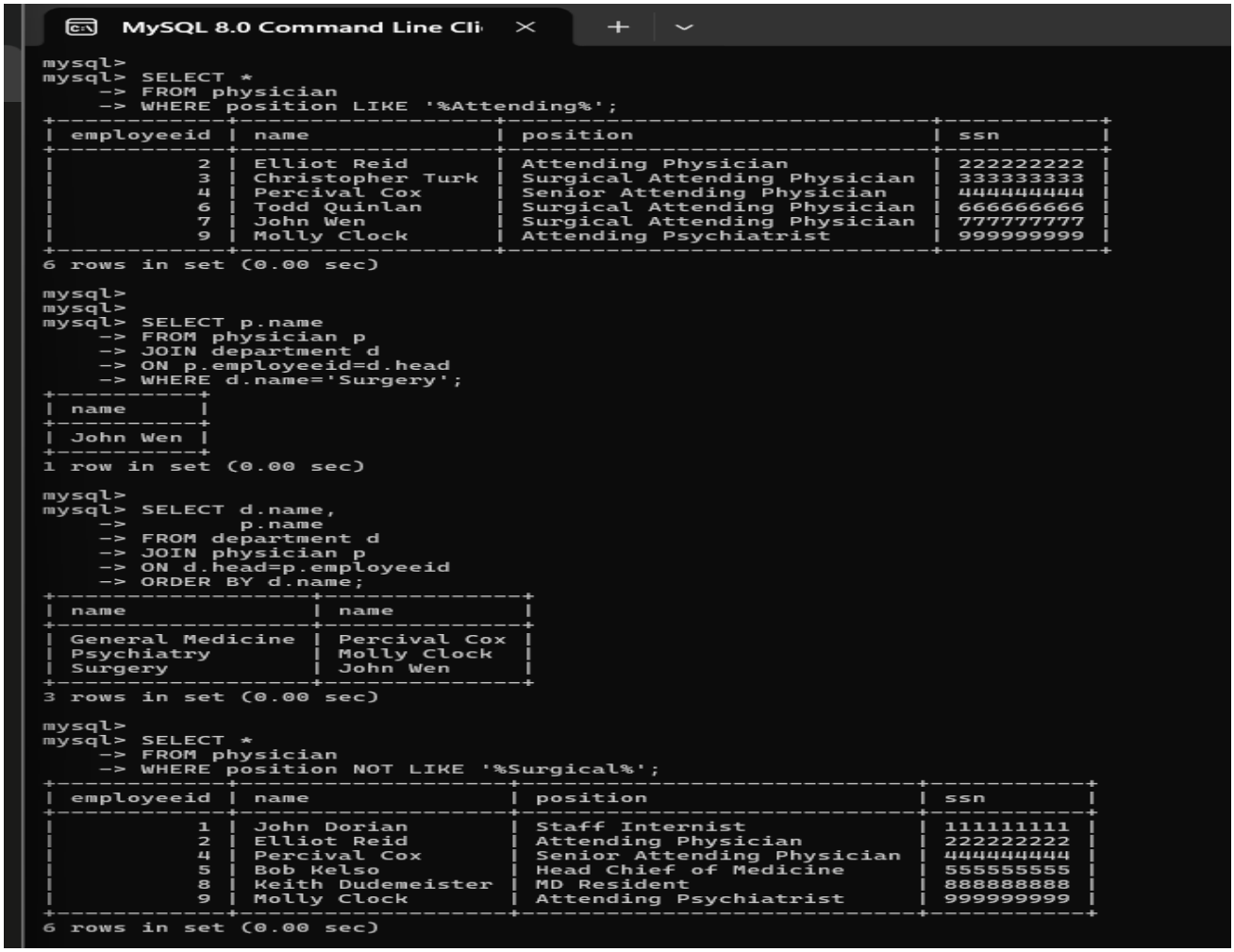
mysql>
mysql> SELECT position,
    ->        COUNT(*) AS Total_Employees
    -> FROM physician
    -> GROUP BY position
    -> ORDER BY Total_Employees DESC;
+-----------------------------+-----------------+
| position                    | Total_Employees |
+-----------------------------+-----------------+
| Surgical Attending Physician|               3 |
| Staff Internist             |               1 |
| Attending Physician         |               1 |
| Senior Attending Physician  |               1 |
| Head Chief of Medicine      |               1 |
| MD Resident                 |               1 |
| Attending Psychiatrist      |               1 |
+-----------------------------+-----------------+
7 rows in set (0.00 sec)
```


Subquery Analysis:

• WHERE employeeid IN (SELECT head FROM department):
  - Department Heads: Percival Cox (4), John Wen (7), Molly Clock (9)
• WHERE employeeid NOT IN (SELECT head FROM department):
  - Non-Department Heads: 6 physicians (John Dorian, Elliot Reid, etc.)
• GROUP BY position ORDER BY Total_Employees DESC:
  - Ranking shows Surgical Attending Physician (3) leads all positions

SECTION 5: Complex JOINs & Pattern Matching

Screenshot 6: LIKE Patterns & JOIN Operations



Advanced Query Results:

• WHERE position LIKE '%Attending%': Found 6 physicians with "Attending" in title
  - Includes Attending Physicians, Surgical Attending, etc.
• JOIN department WHERE d.name = 'Surgery':
  - Returns John Wen as Surgery department head
• JOIN + ORDER BY department name:
  - General Medicine (Percival Cox), Psychiatry (Molly Clock), Surgery (John Wen)
• WHERE position NOT LIKE '%Surgical%':
  - Non-surgical physicians: 6 total (excluding surgical roles)

COMPREHENSIVE QUERY REFERENCE

Query Type	Description & Purpose	Result / Output
INNER JOIN	Link departments with their heads	3 departments with assigned heads
WHERE	Filter by position or criteria	3 Surgical Attending Physicians
LIKE	Pattern matching in strings	Names starting with John: 2 results
COUNT	Total number of records	Total Physicians: 9
COUNT DISTINCT	Unique values in column	Distinct Positions: 6
GROUP BY	Aggregate data by category	Position distribution: 7 groups
HAVING	Filter grouped results	Positions with 2+ employees: 1
ORDER BY	Sort results ASC or DESC	Alphabetical or by employee ID
SUBQUERY	Nested query for filtering	Department heads: 3, Non-heads: 6
EXISTS	Check if records exist	All 3 departments have valid heads

Key Statistics & Findings

✓ Total Physicians: 9 medical professionals
✓ Total Departments: 3 (General Medicine, Surgery, Psychiatry)
✓ Unique Positions: 6 different roles in hospital
✓ Most Common Position: Surgical Attending Physician (3 employees)
✓ Department Heads: 3 specialized physicians (Percival Cox, John Wen, Molly Clock)
✓ Non-Department Heads: 6 physicians in regular positions
✓ All queries executed successfully with accurate results